PhytoScope One local framework for diverse plant genomes

Public files → a shared locus model → inspectable, exportable and recoverable results

a Species as data

Crop Medicinal Ornamental

Compatible public inputs

Annotation
GFF3 / GTF

Genome sequence
Matching FASTA

Expression
Matrix + sample labels

b Shared local model

One import / query code path

Sequence IDs and coordinates

Gene / locus

Transcript 1 | Transcript 2

Exon • CDS • UTR relationships
Source features / inference labels

Save ↔ reopen project
Inputs, locus and results

c Locus-centered queries and outputs

Structure *V. umbellata · LOC124833352*

NC_142174.1 | chromosome 10 | minus strand

16,647 20,000 23,821

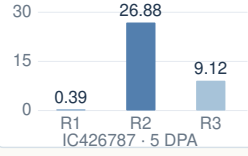
XM_088100767.1 CDS Inferred UTR

Sequence export
LOC124833352

Transcript	2,060 nt
CDS	1,983 nt
Protein	660 aa

>CDS ATGGATTATGATCG…

Expression display
LOC148191553 · gene TPM



Checked against independent source-derived references

d Public-data evidence

7 plant species evaluated

Structural fidelity

4,641,758 recognized tuples

Arabidopsis · bundled	811,827
Arabidopsis · full	791,557
Maize · excerpt	377
Maize · full	1,161,691
Rice · excerpt	347
Rice · full	550,921
Soybean · excerpt	839
Soybean · full	1,324,199

8 inventories · 0 missing · 0 extra

Sequence reconstruction

15 loci 25 coding transcripts

Plant	tx	Span	RNA	CDS
Vigna	6	✓	✓	✓
Andrographis	14	✓	✓	✓
Hibiscus	5	✓	✓	✓

5 selected coding loci per plant
Strand-aware sequence SHA-256 checks

Expression ID reconciliation

Public reads → exact RefSeq RNA → gene TPM

100%

34,914 / 34,914
gene IDs matched
0 unmatched IDs

Reads IC426787 ≠ reference VRB3
Supports import / ID compatibility; genotype-matched biology is not tested.

Versioned public inputs → **Source manifests + checksums** → **Independent checks** → **Reproducible scripts and saved projects**